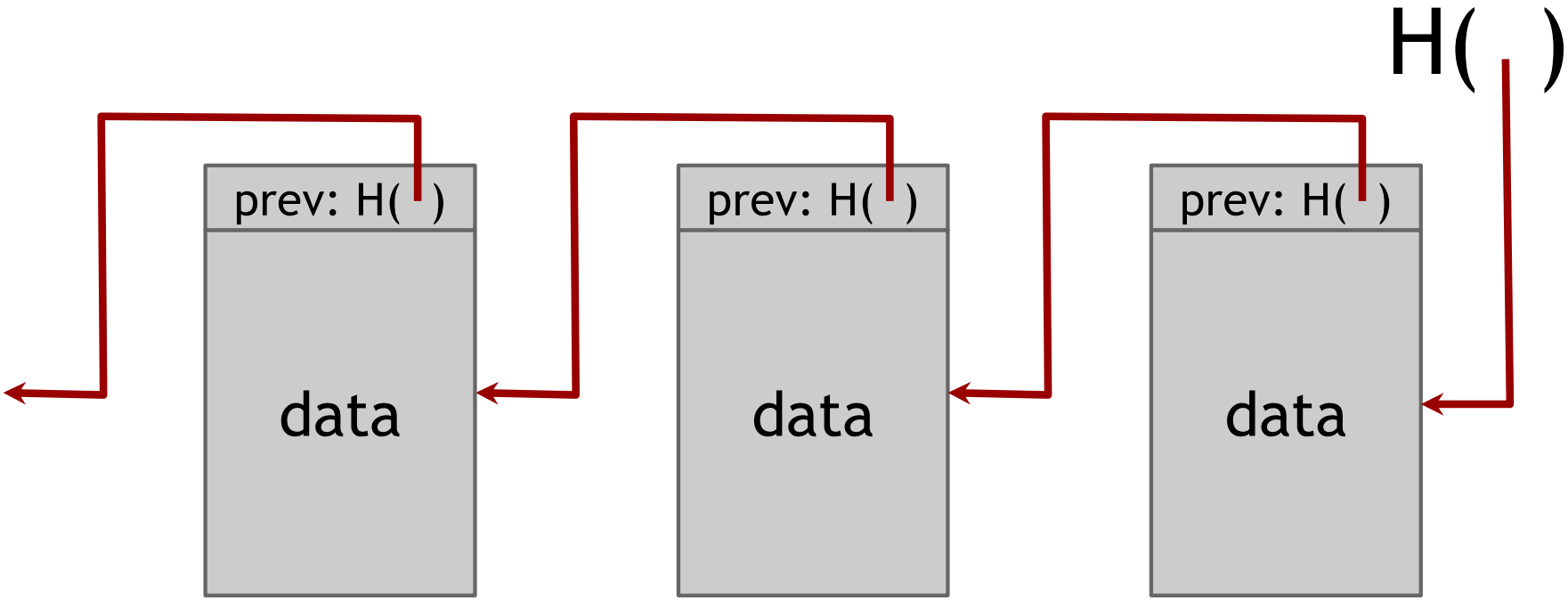
Intro to Blockchain and Cryptocurrencies

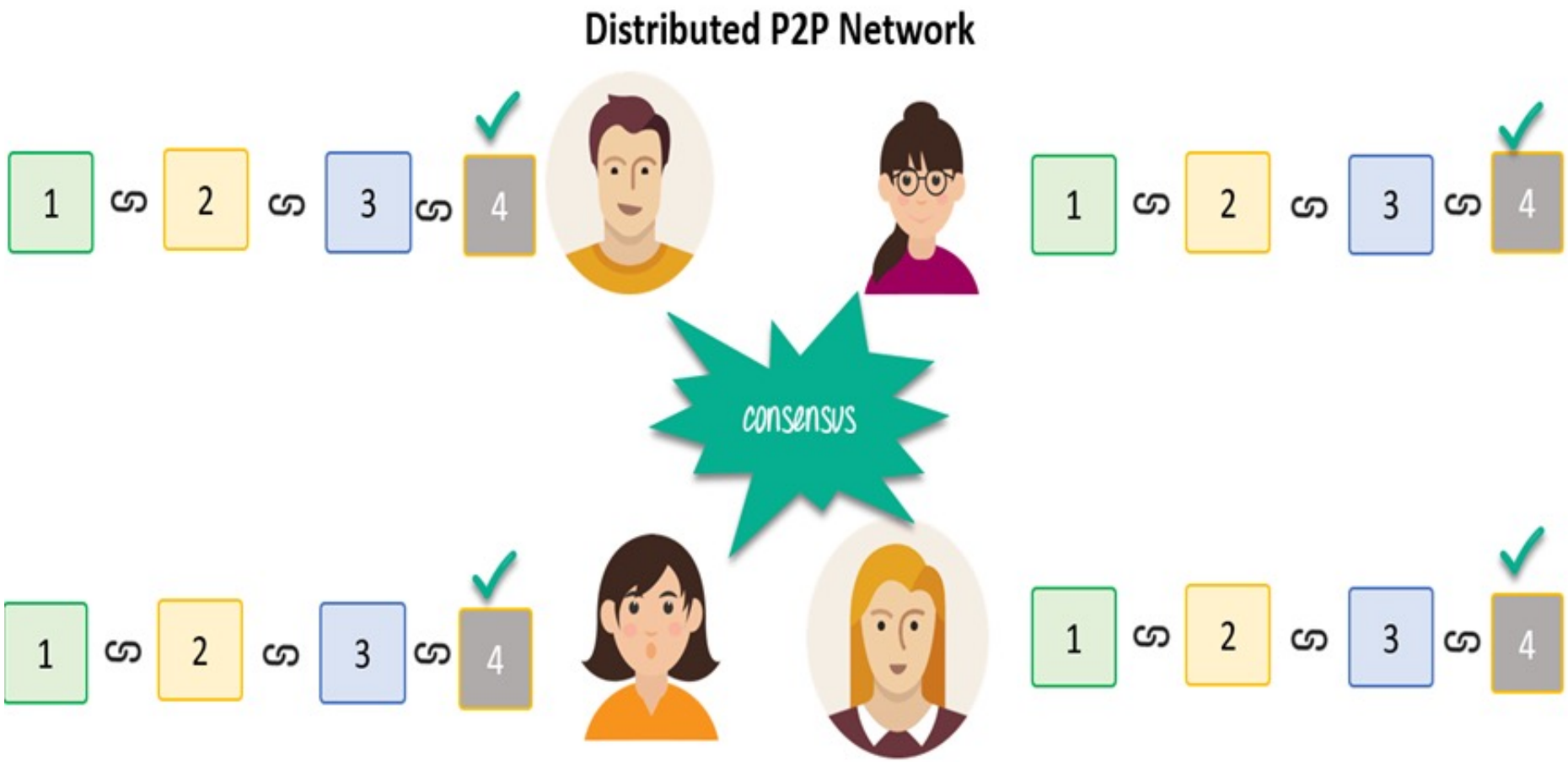
Slides by Arvind Narayanan et al.

linked list with hash pointers = “block chain”

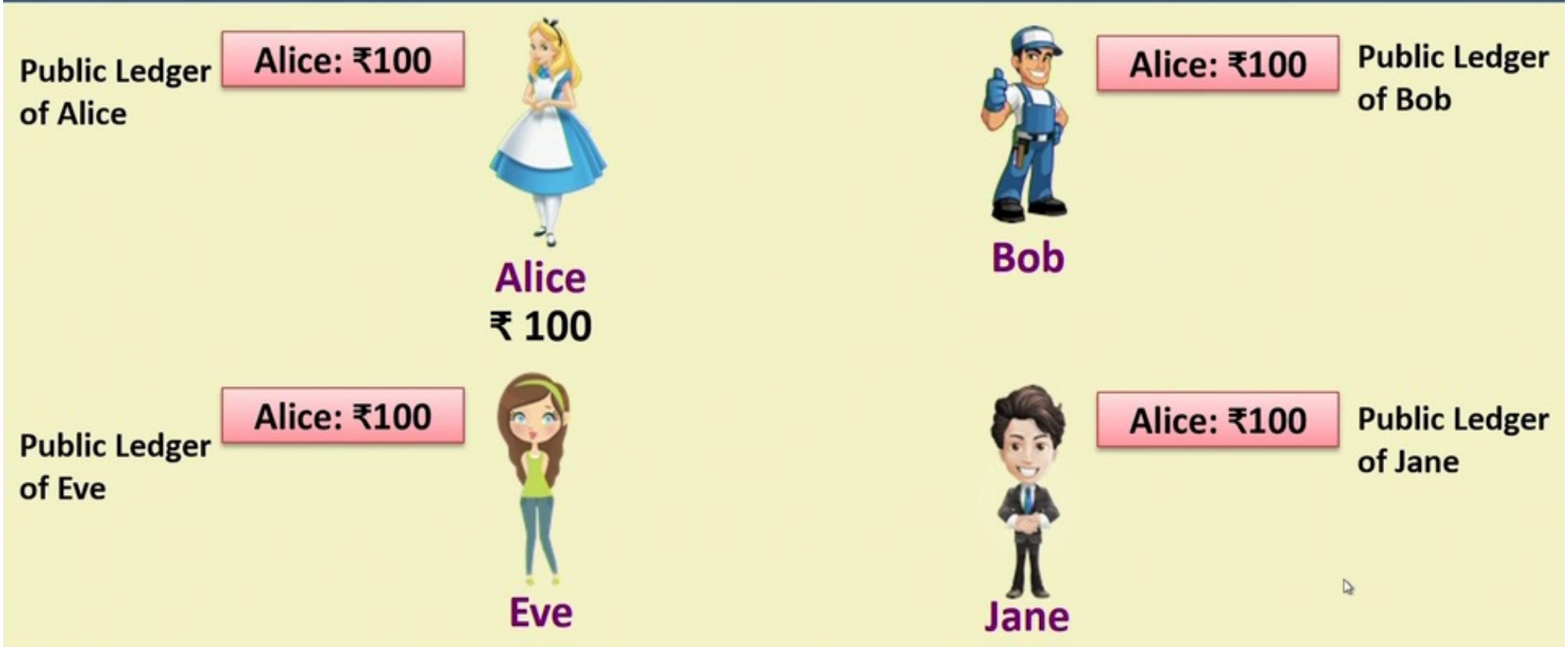


use case: tamper-evident log

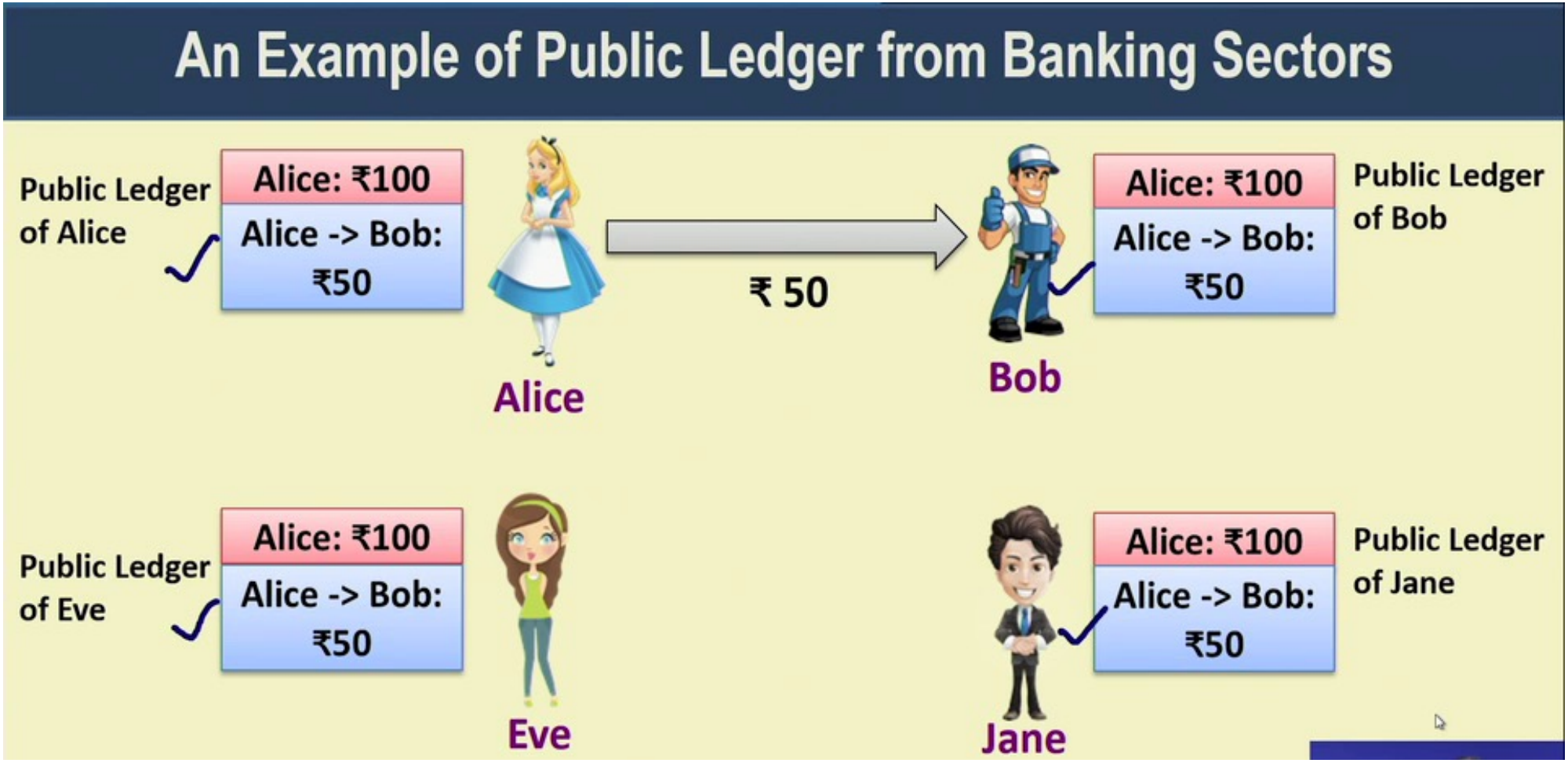
How to achieve consistency?



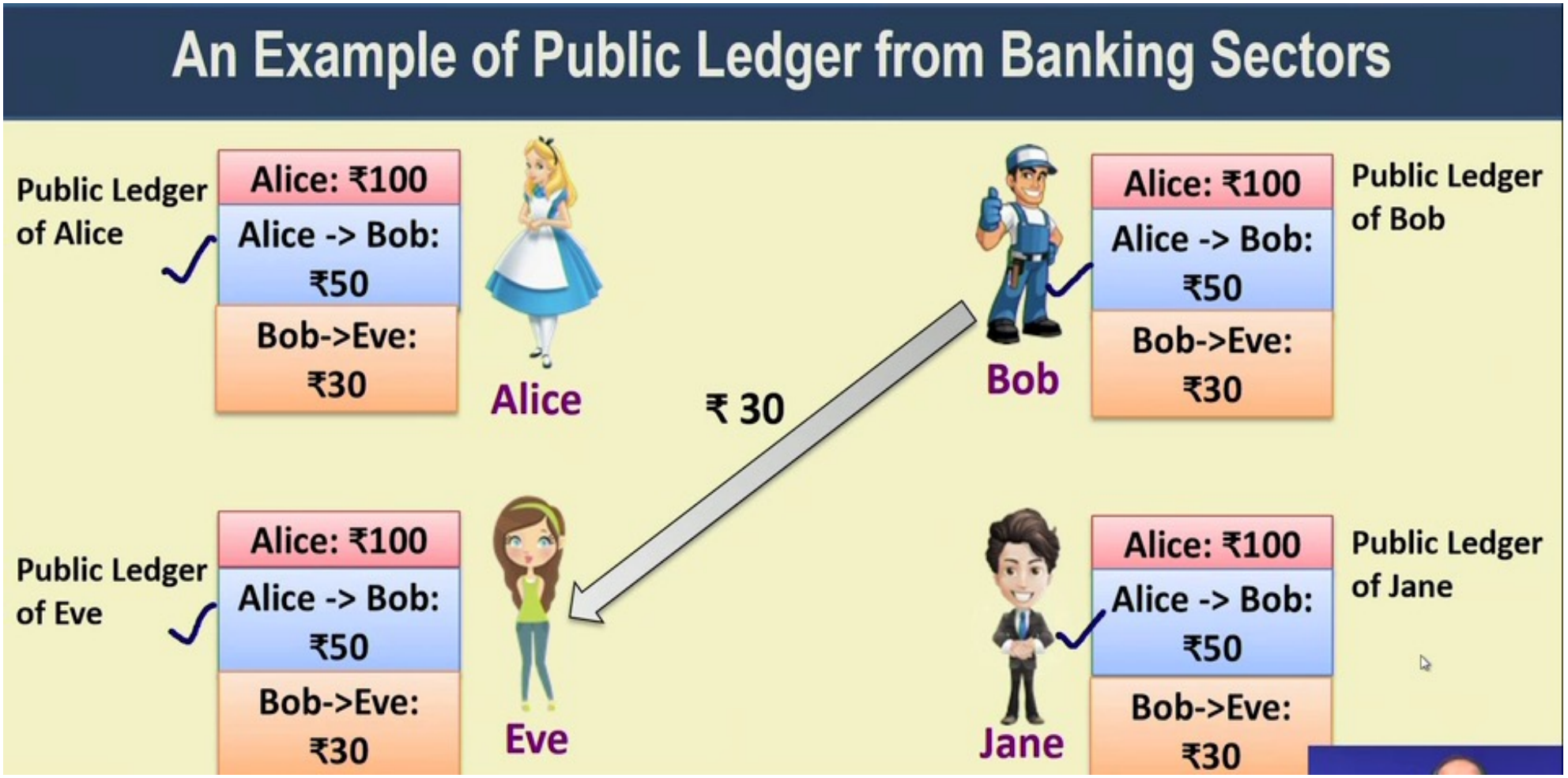
An Example of Public Ledger from Banking Sectors



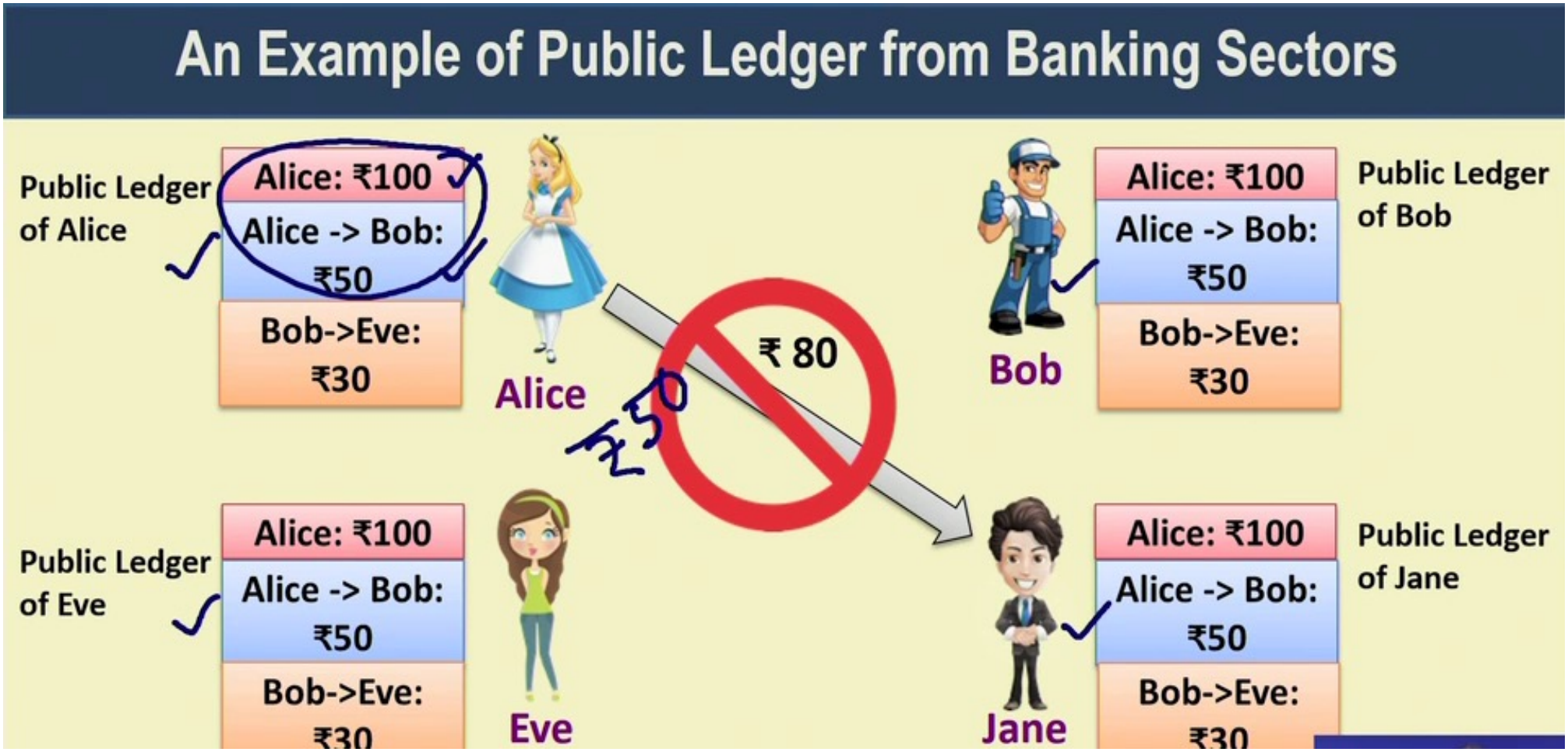
Courtesy: Lectures by Dr. Sandip Chakraborty



Courtesy: Lectures by Dr. Sandip Chakraborty



Courtesy: Lectures by Dr. Sandip Chakraborty



Courtesy: Lectures by Dr. Sandip Chakraborty

The path to decentralization

- technology & incentive design

All Participants

Who maintains the ledger of transactions? (and how?)

Consensus

All Participants

Who determines the validity of transactions to be included in the ledger?

Bitcoin Script

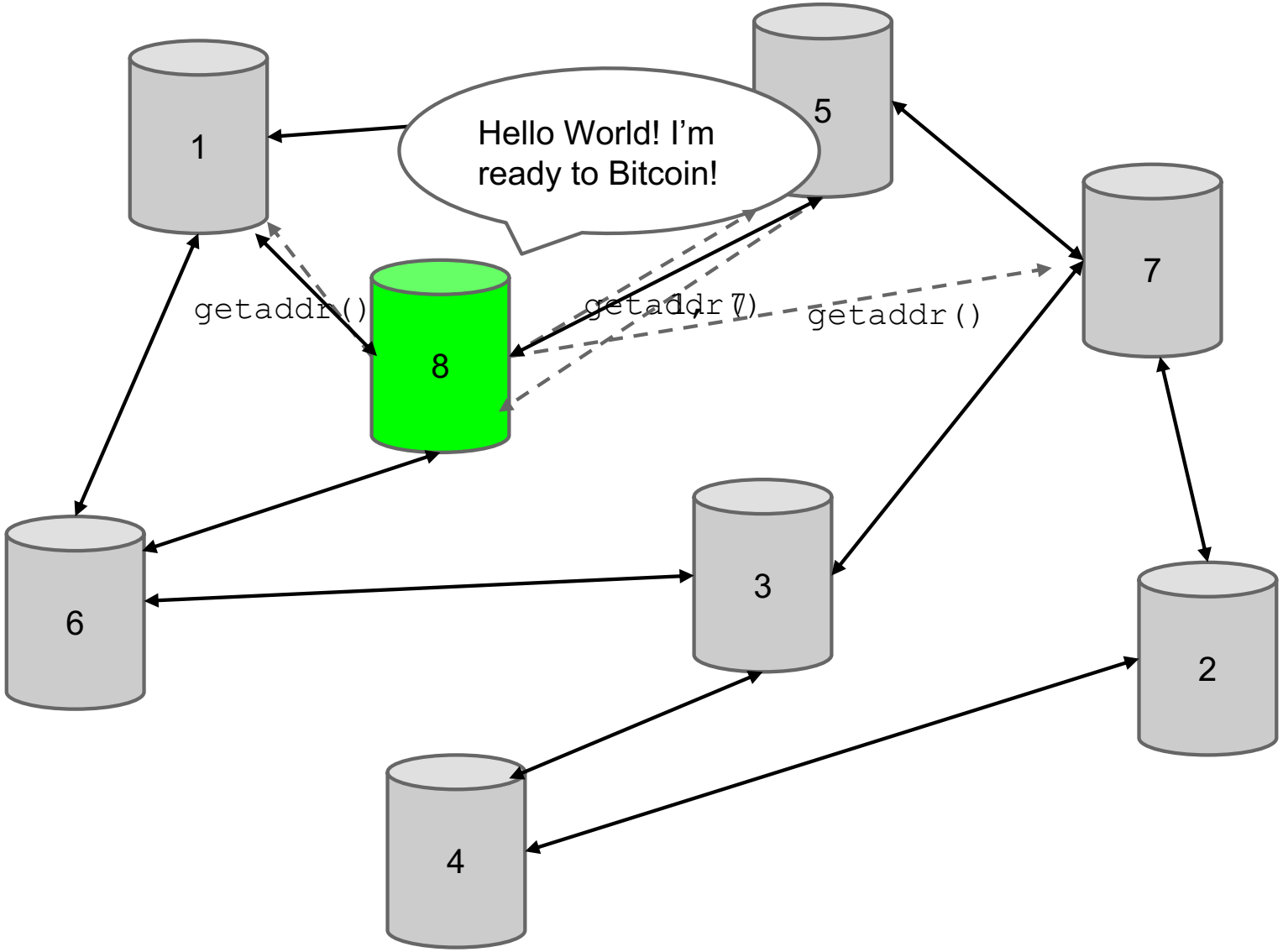
Who creates new Bitcoins?

Reward for Mining

Bitcoin P2P network

- Ad-hoc protocol (runs on TCP port 8333)
- Ad-hoc network with random topology
- All nodes are equal
- New nodes can join at any time
  - Network Changes over time - dynamic
- No explict way to leave network
  - Forget non-responding nodes after 3 hr

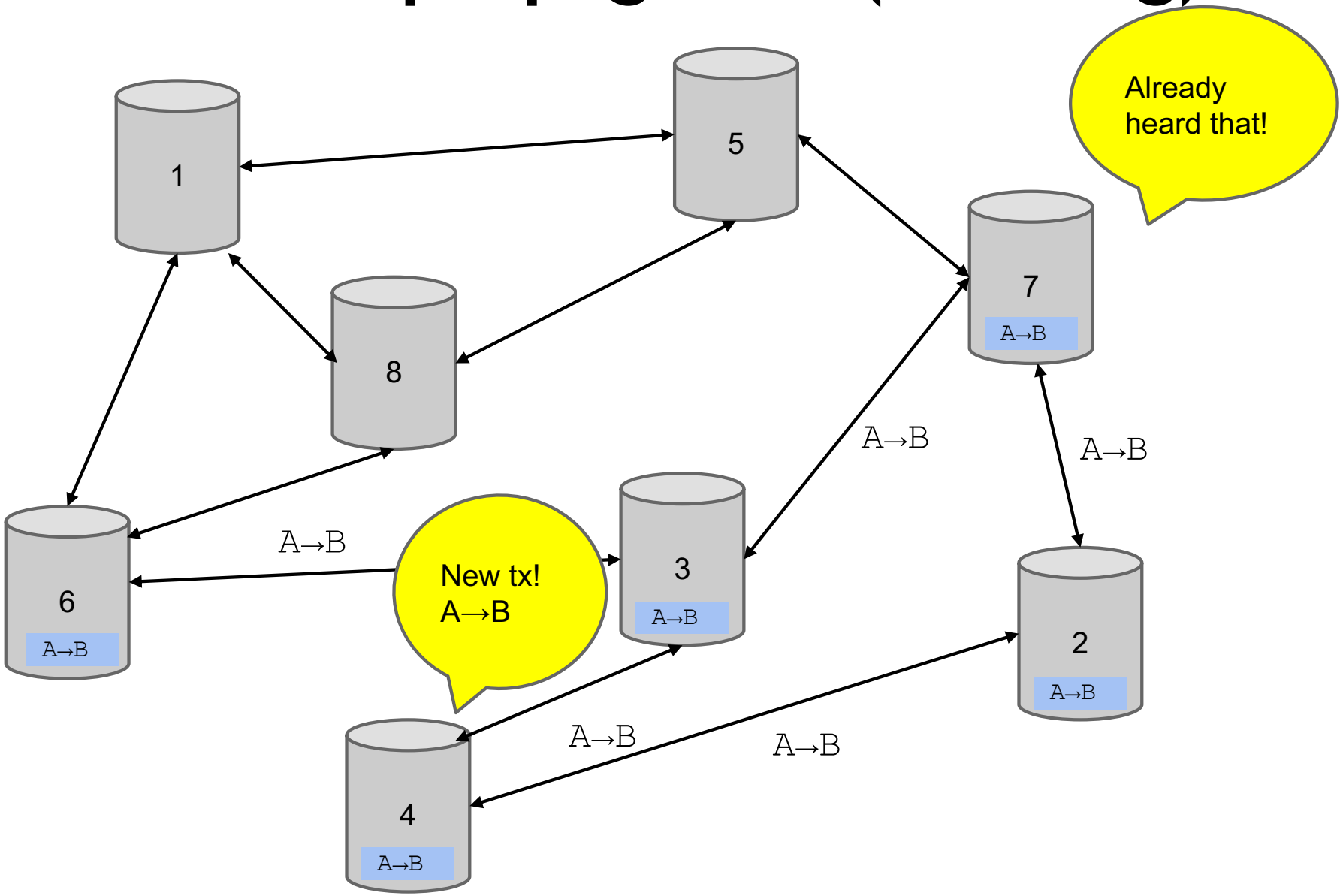
Joining the Bitcoin P2P network



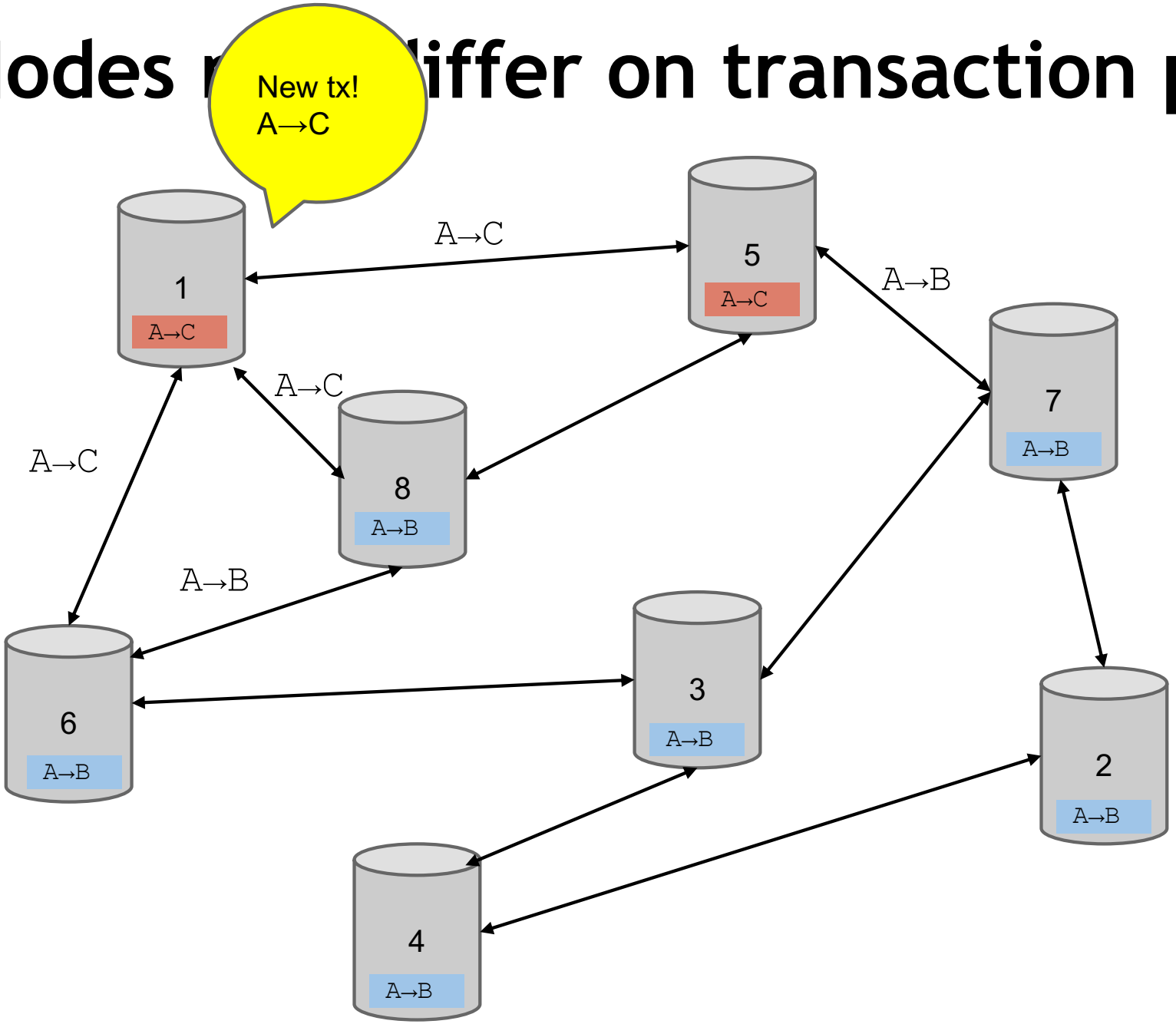
BLOCKCHAIN WORKING PRINCIPLE

HOW THE BLOCKCHAIN WORKS?

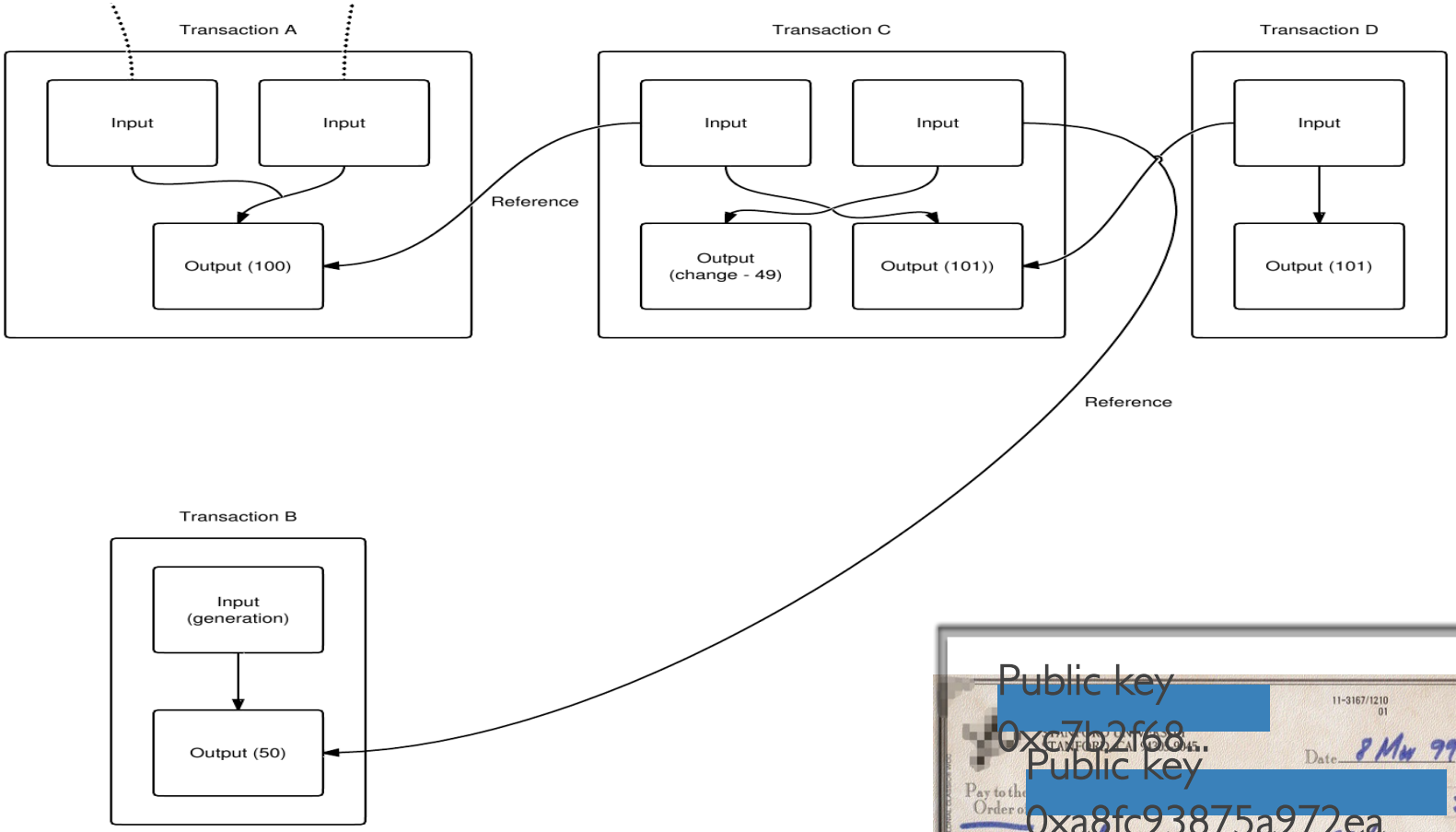
Transaction propagation (flooding)

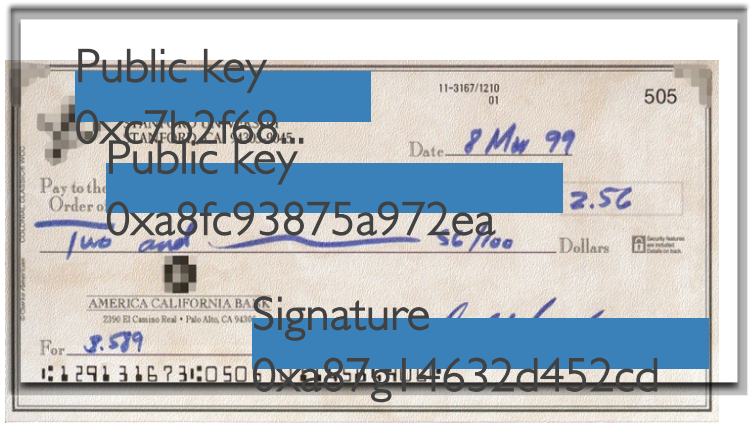


Nodes may differ on transaction pool



Bitcoin Transaction Structure





The real deal: a Bitcoin transaction

metadata

```
{
  "hash":"5a42590fbe0a90ee8e8747244d6c84f0db1a3a24e8f1b95b10c9e050990b8b6b",
  "ver":1,
  "vin_sz":2,
  "vout_sz":1,
  "lock_time":0,
  "size":404,
  "in":[
```

input(s)

```
    {
      "prev_out":{
        "hash":"3be4ac9728a0823cf5e2deb2e86fc0bd2aa503a91d307b42ba76117d79280260",
        "n":0
      },
      "scriptSig":"30440..."
    },
    {
      "prev_out":{
        "hash":"7508e6ab259b4df0fd5147bab0c949d81473db4518f81afc5c3f52f91ff6b34e",
        "n":0
      },
      "scriptSig":"3f3a4ce81...."
    }
  ],
```

Owner's Signature

Owner's Signature

output(s)

Recipient

```
  "out":[
    {
      "value":"10.12287097",
      "scriptPubKey":"OP_DUP OP_HASH160 69e02e18b5705a05dd6b28ed517716c894b3d42e OP_EQUALVERIFY OP_CHECKSIG"
    }
  ]
}
```

Alice → Bob

Transaction Created by Alice and already in Blockchain

metadata

input(s)

output(s) 2


```
{
  "hash":"5a42590fbe0a90ee8e8747244d6c84f0db1a3a24e8f1b95b10c9e050990b8b6b",
  "ver":1,
  "vin_sz":2,
  "vout_sz":1,
  "lock_time":0,
  "size":404,
  "in":[
    {
      "prev_out":{
        "hash":"3be4ac9728a0823cf5e2deb2e86fc0bd2aa503a91d307b42ba76117d79280260",
        "n":0
      },
        "scriptSig":"30440..."
    },
    {
      "prev_out":{
        "hash":"7508e6ab259b4df0fd5147bab0c949d81473db4518f81afc5c3f52f91ff6b34e",
        "n":0
      },
      "scriptSig":"3f3a4ce81...."
    }
  ],
  "out":[
    {
      "value":"10.12287097",
      "scriptPubKey":"OP_DUP OP_HASH160 69e02e18b5705a05dd6b28ed517716c894b3d42e OP_EQUALVERIFY OP_CHECKSIG"
    }
  ]
```


Signed by Alice

Recipient is Bob

Bob → Carol

metadata

input(s)

output(s)

1

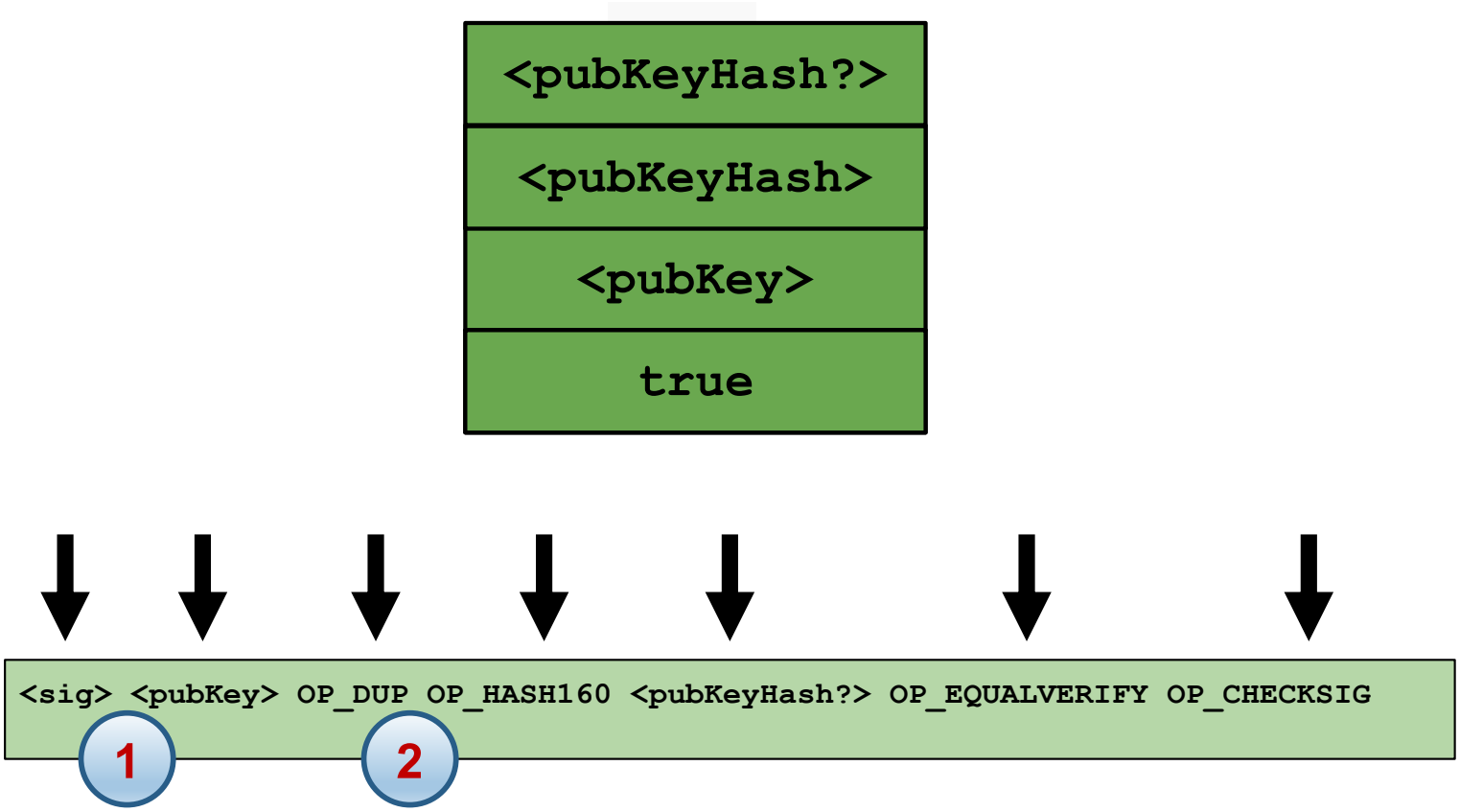

```
{
  "hash":"5a42590fbe0a90ee8e8747244d6c84f0db1a3a24e8f1b95b10c9e050990b8b6b",
  "ver":1,
  "vin_sz":2,
  "vout_sz":1,
  "lock_time":0,
  "size":404,
  "in":[
    {
      "prev_out":{
        "hash":"3be4ac9728a0823cf5e2deb2e86fc0bd2aa503a91d307b42ba76117d79280260",
        "n":0
      },
        "scriptSig":"30440..."
    },
    {
      "prev_out":{
        "hash":"7508e6ab259b4df0fd5147bab0c949d81473db4518f81afc5c3f52f91ff6b34e",
        "n":0
      },
      "scriptSig":"3f3a4ce81...."
    }
  ],
  "out":[
    {
      "value":"10.12287097",
      "scriptPubKey":"OP_DUP OP_HASH160 69e02e18b5705a05dd6b28ed517716c894b3d42e OP_EQUALVERIFY OP_CHECKSIG"
    }
  ]
```


Signed by Bob

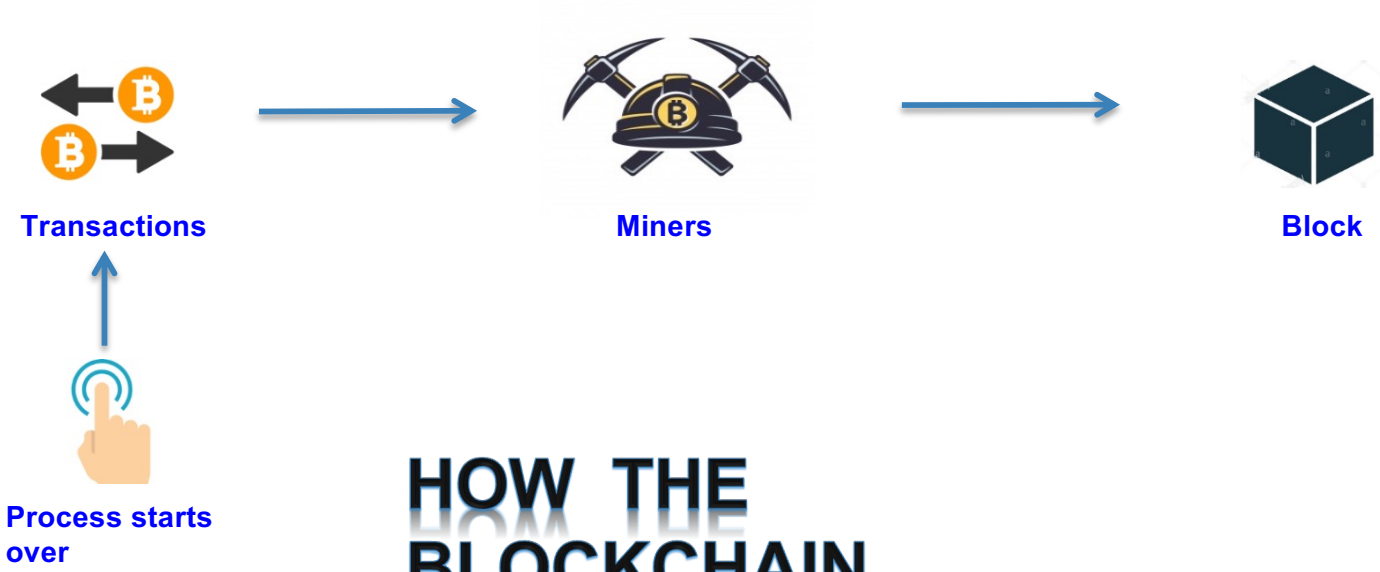
Recipient is Carol

New Transaction created by Bob

Bitcoin script execution example

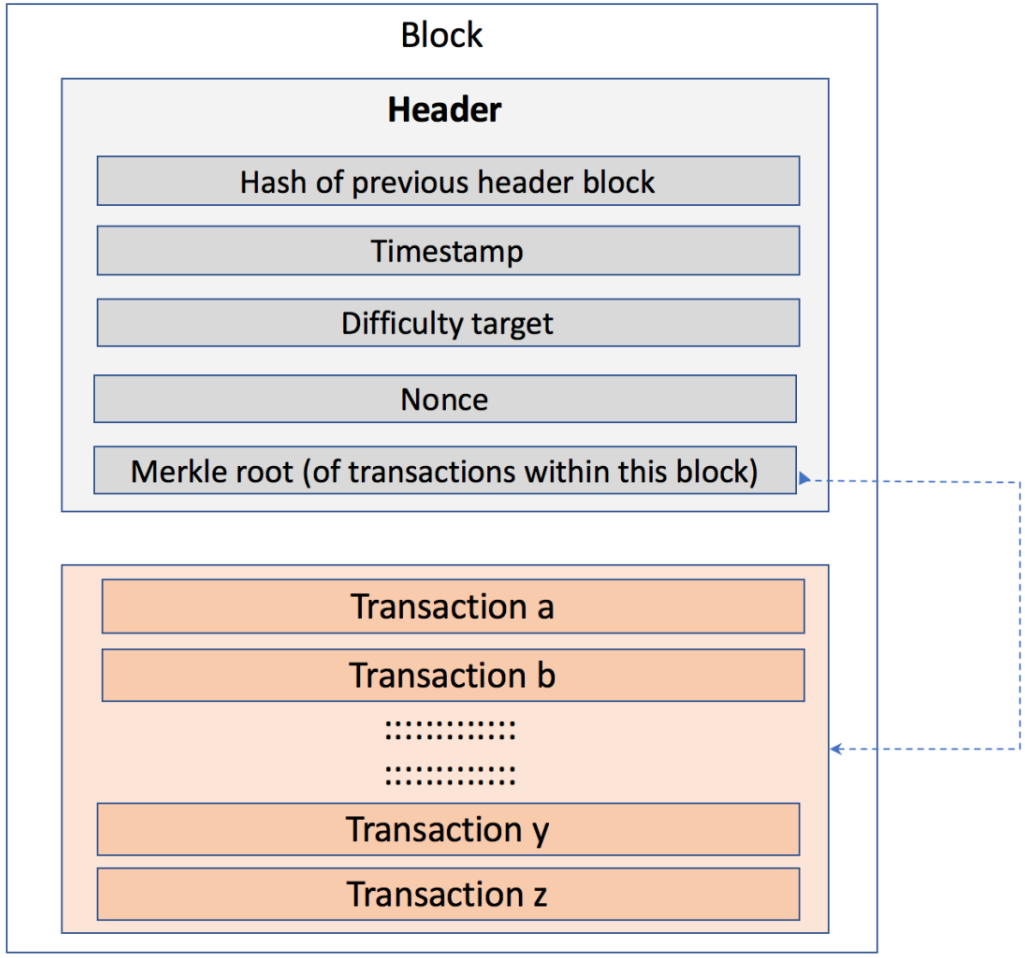


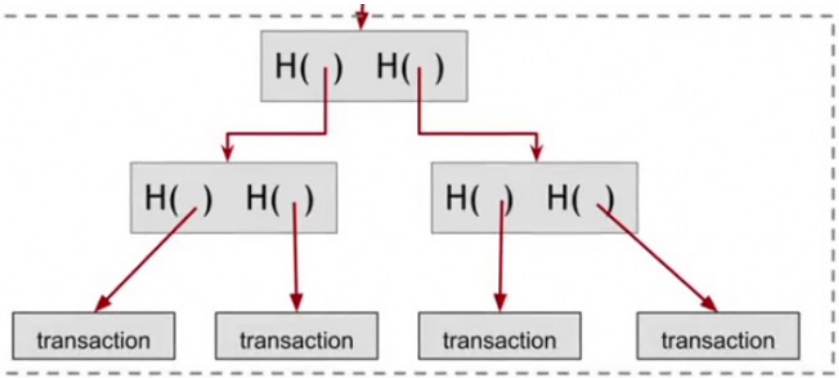
BLOCKCHAIN WORKING PRINCIPLE



HOW THE BLOCKCHAIN WORKS?

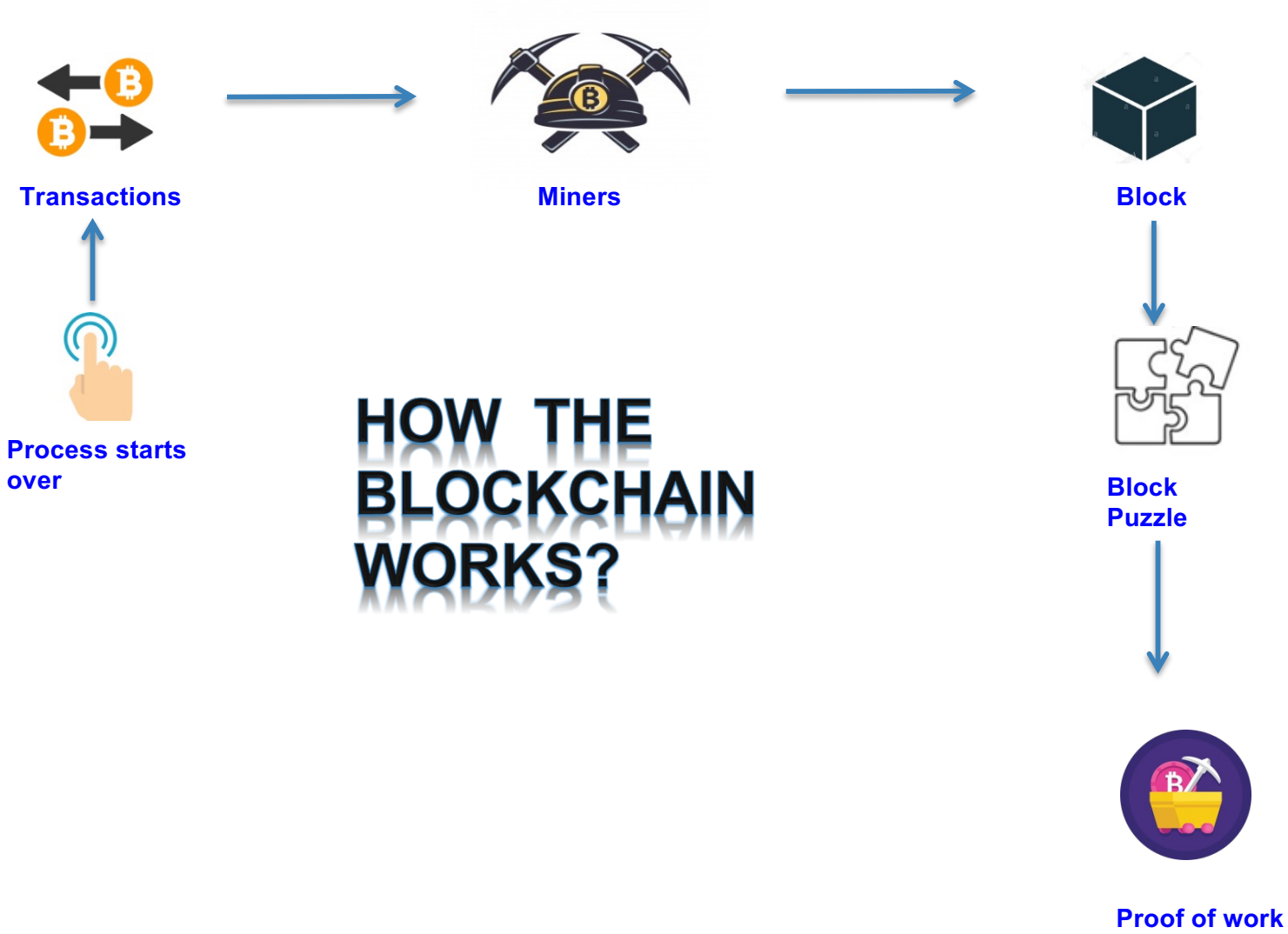
Block Structure



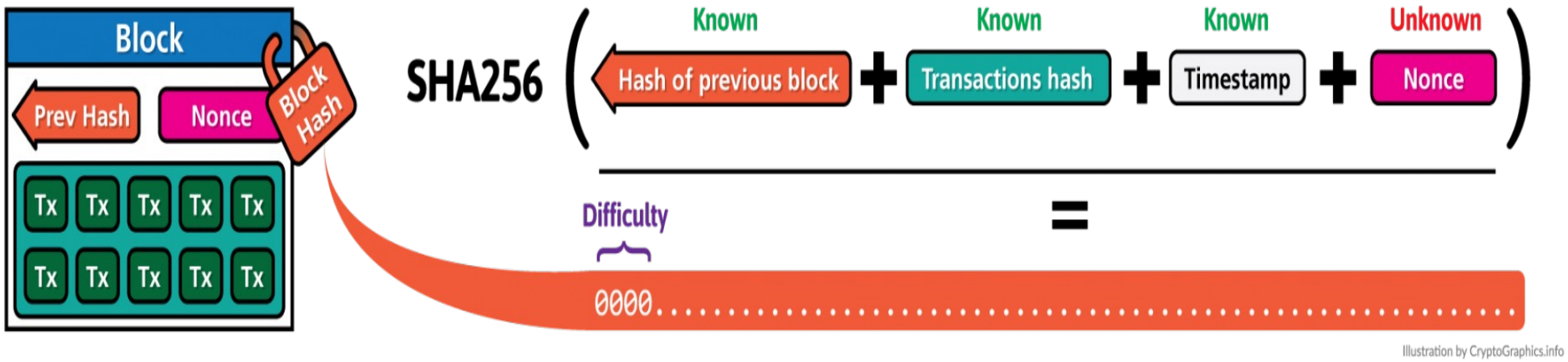


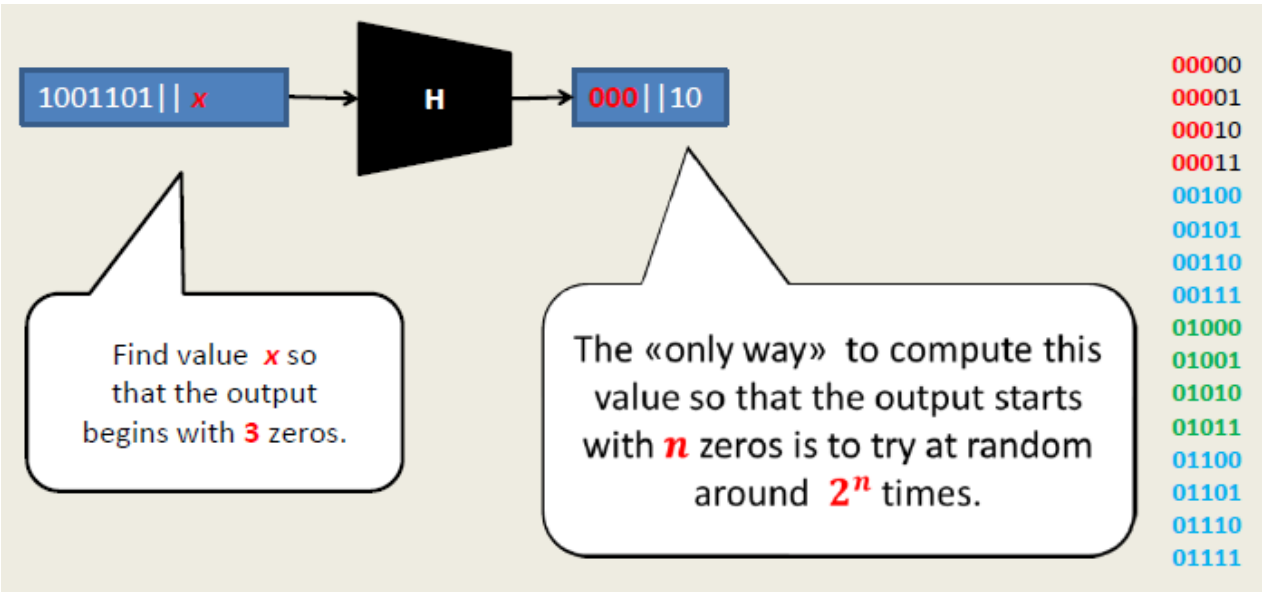
Hash tree (Merkle tree)

BLOCKCHAIN WORKING PRINCIPLE



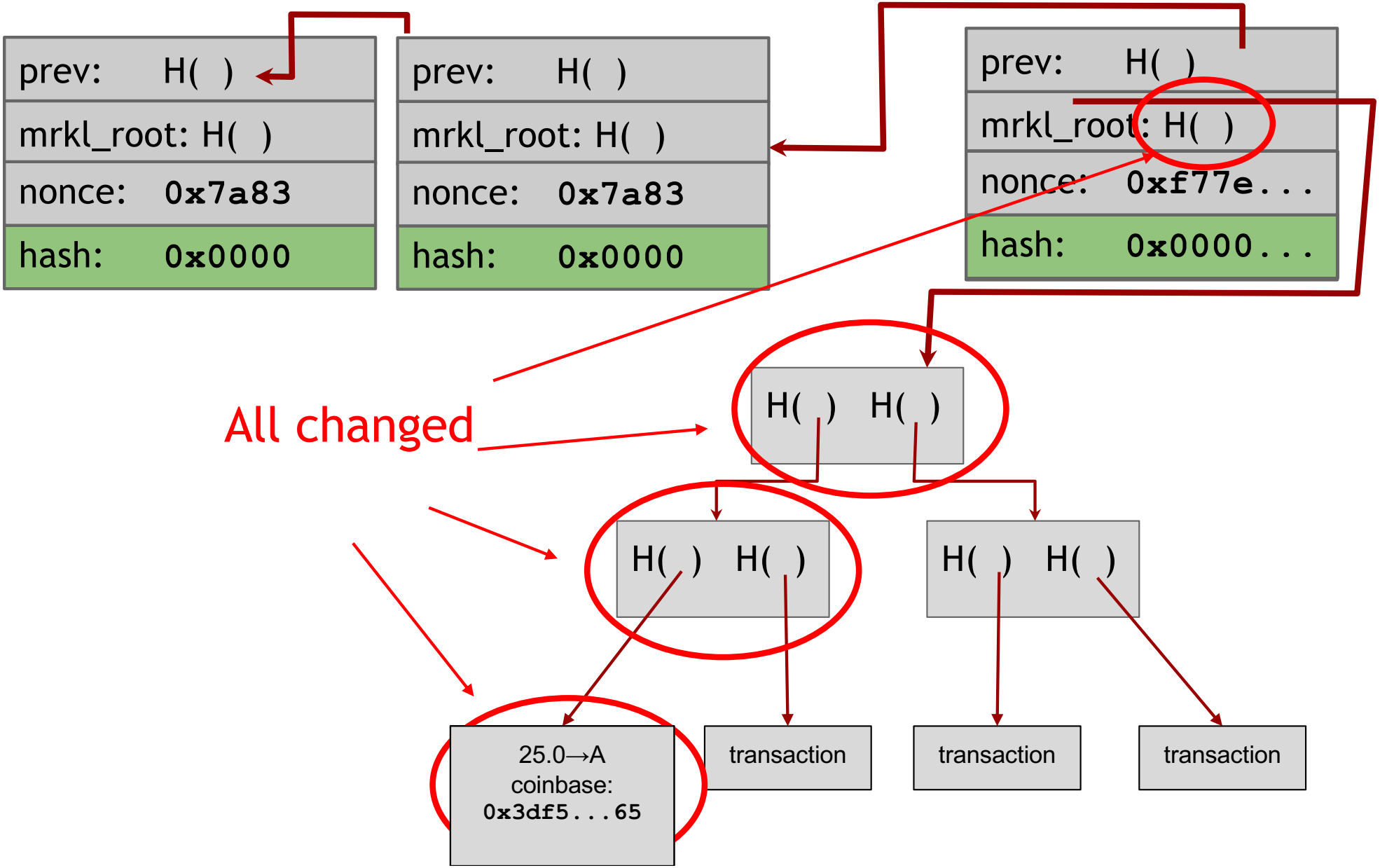
Proof of Work





Proof of Work **[Back2002]**

Mining a block: Proof-of-Work



Forbes

Billionaires Innovation Leadership Money Consumer Industry

Ad closed by Google
Report this ad Why this ad?

EDITOR'S PICK | 23,388 views | Apr 19, 2018, 11:09pm

Bitcoin's Energy Consumption Can Power An Entire Country -- But EOS Is Trying To Fix That

Sherman Lee Contributor
I write about deep tech, crypto, and artificial intelligence.

Back to Articles

Bitcoins Energy Consumption An Unsustainable Protocol That Must Evolve?

3

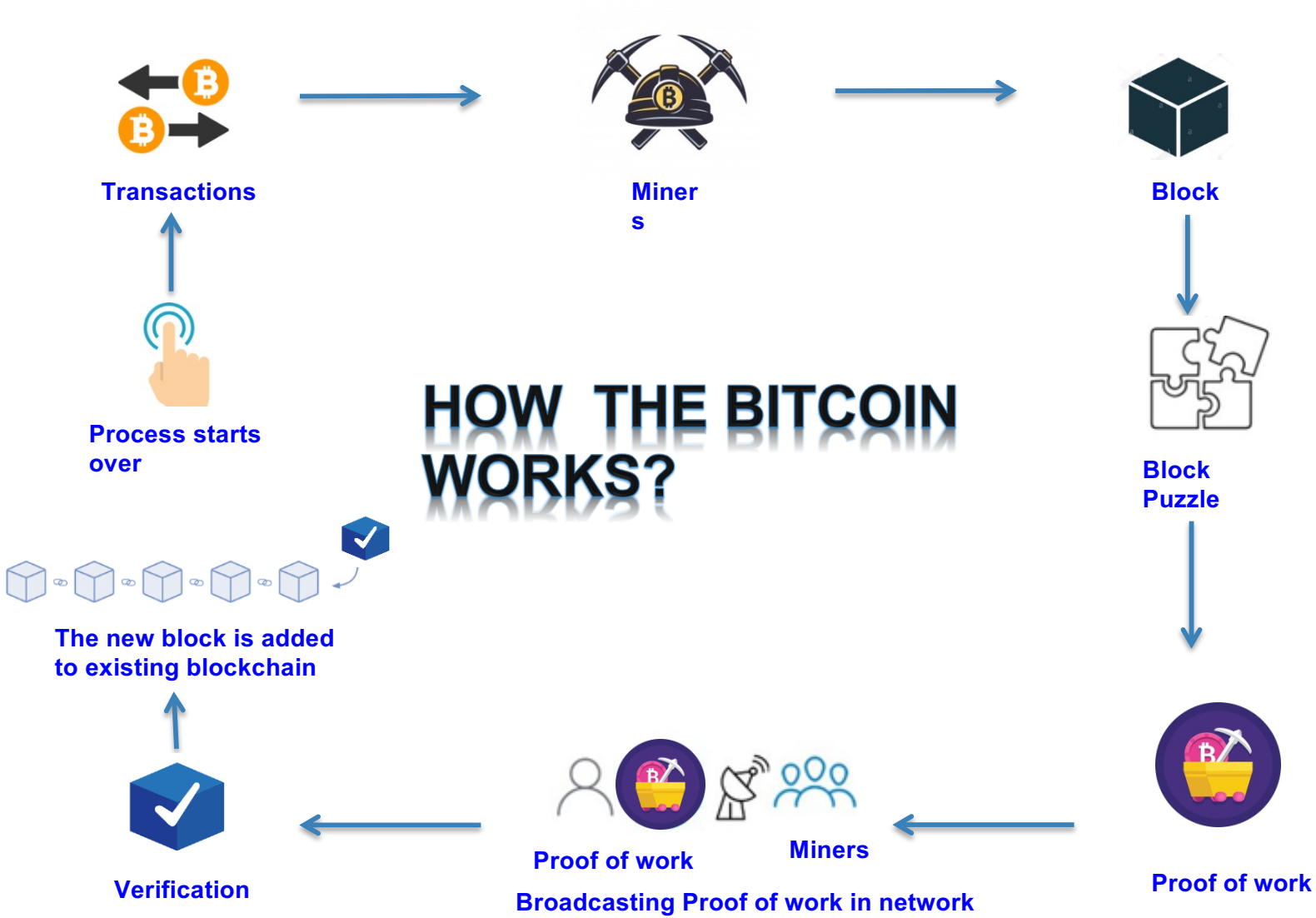
3

By john lilic #Blockchain 101 #Blockchain for Business #Blockchain for Investors

Alternative Consensus

- Proof of Stake
- PBFT
- Raft

BITCOIN WORKING PRINCIPLE



Mining Bitcoins in 6 easy steps

1. Join the network, listen for transactions
    a. Validate all proposed transactions
2. Miners will assemble them into new blocks and solve puzzle
3. On success, broadcast the new blocks
4. Listen for new blocks, maintain block chain
    a. When a new block is proposed, validate it
5. Find the nonce to make your block valid
6. Hope everybody accepts your new block
7. Profit!

Rewards and Transaction Fees!